

Introduction to CUDA, Part I

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https://github.com/zuzanna-m-j/IntroToCUDA_S2026.git

What is CUDA

- (Compute Unified Device Architecture)
- Parallel programming platform for NVIDIA GPUs
- Thread-based programming model
- Not a programming language but provides language extensions for C++, Fortran and Python
- Software developed to match the hardware

People: HTML is not a real programming language

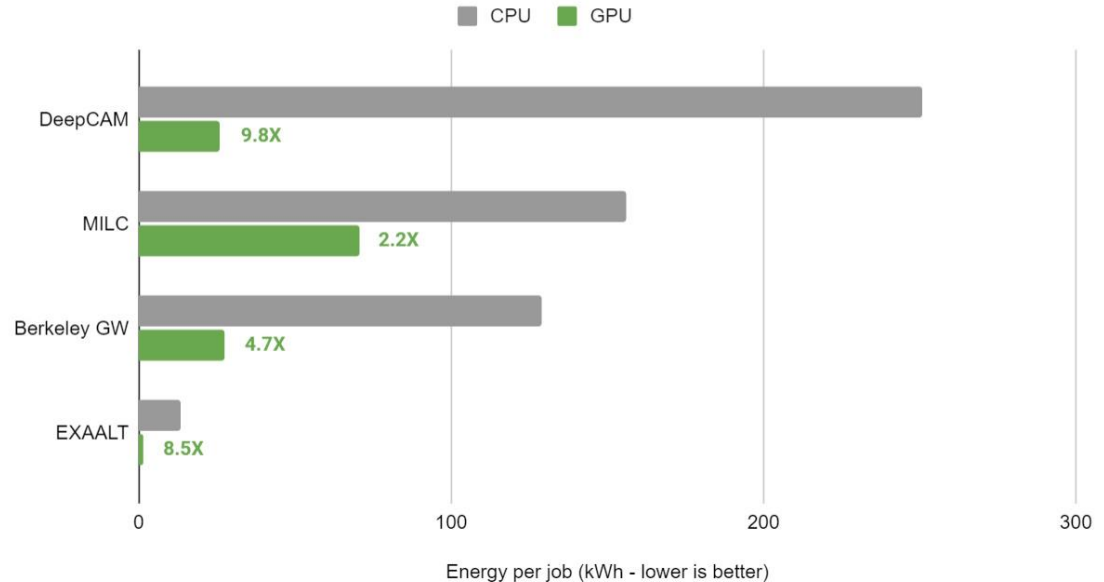
CUDA:



Why use a GPU?

- Efficiency and performance
- CPU limiting factors:
 - (Mostly) Serial execution
 - Moore's law limit
 - Clock speed
- Benefits of using GPUs:
 - Massive parallelism
 - Energy efficiency
 - Python is now mainstream and everyone “knows it”.
But understanding CUDA still gives you the bragging rights.

Energy Consumed per Job



<https://blogs.nvidia.com/blog/gpu-energy-efficiency-nersc/>

GPUs in scientific computing

- Machine Learning
- Molecular Dynamics
- Fluid Dynamics
- Quantum Mechanics
- Scientific Visualization

GPU is not a silver bullet

CPU programming is like using a handsaw to cut wood: straightforward, might be slower but eventually it will do its job. GPU programming is like using a chain saw: it will cut wood faster if you know what you're doing, might cut your fingers faster if you don't.

Yes!

- Computation intensive tasks (high compute to data load ratio)
- Data accessed in large continuous chunks
- Matrix operations

No!

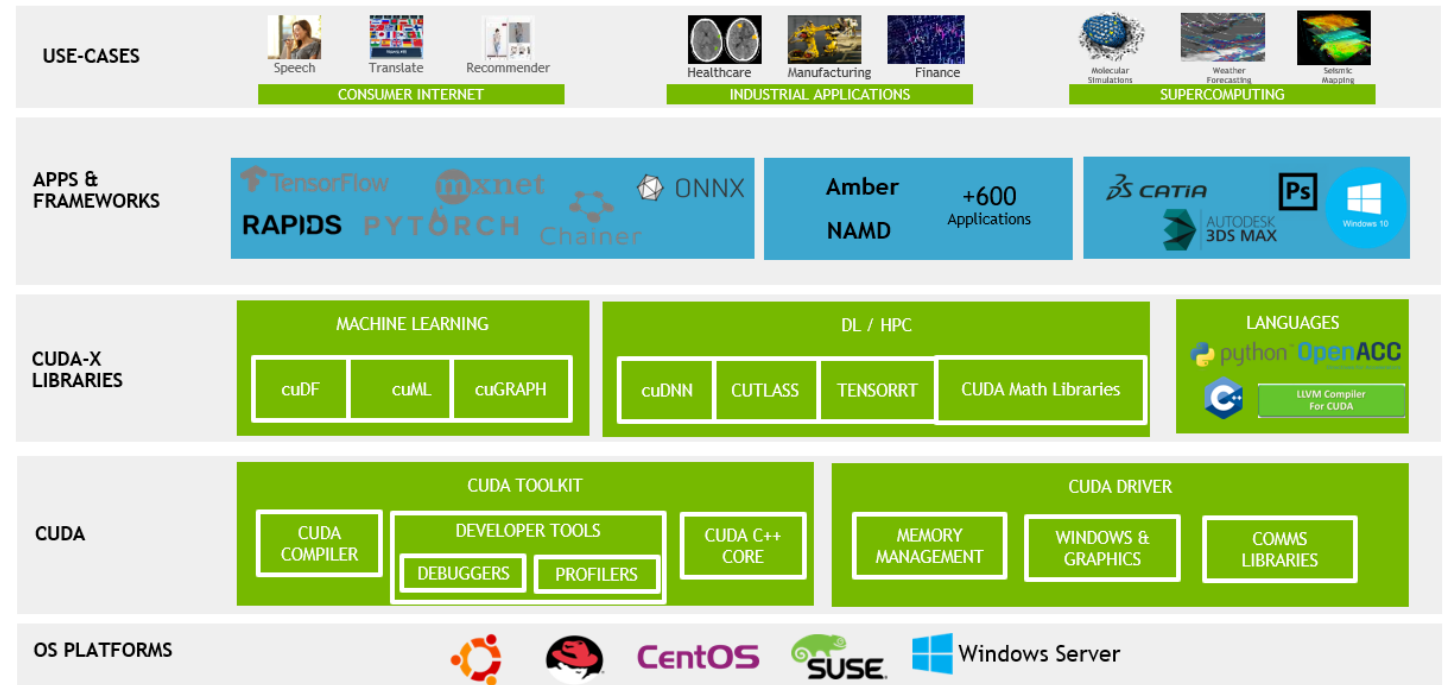
- Code with a lot of branching instructions
- Lots of interdependence between threads
- Processing done on small batches of data
- Irregular and scattered data access pattern

Ways to program a GPU

- GPU programming “languages”:
 - CUDA for NVIDIA (C/C++ and Fortran)
 - HIP for AMD (C only)
- Pragma (compiler directive) based device offloading:
 - OpenMP
 - OpenACC

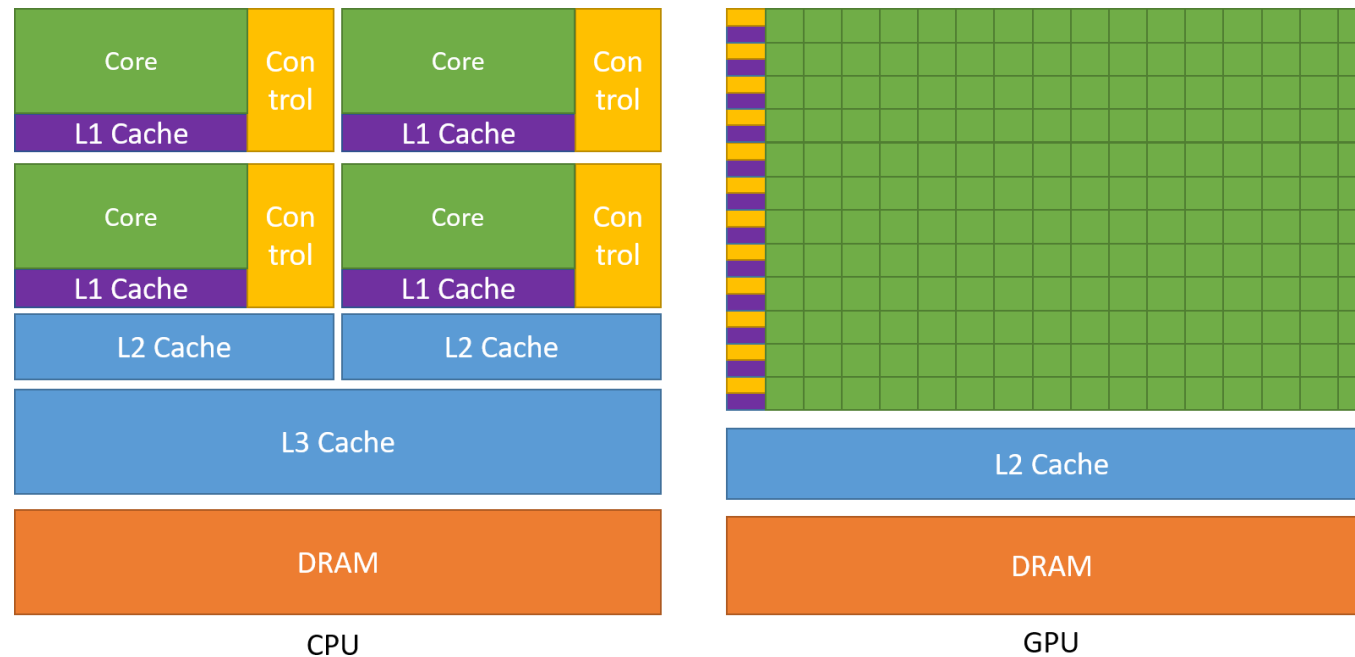
What's in the CUDA toolbox

- CUDA: C/C++ and Fortran
- Compilers: nvcc, nvc, nvc++, nvfortran
- Profiling tools: Nsight Systems, Nsight Compute
- Specialized libraries
 - Math: cuFFT, cuBLAS
 - Algorithms: Thrust
 - Deep Learning: cuDNN
- CUDA-aware MPI



CPU vs GPU

CPU	GPU
Low core count	Very high (1,000s) core count
Low latency	High latency but high throughput
Good for serial execution, can be parallel	Optimized for parallel execution
Fast context switching, branching	Good for SIMD, avoid branching



From NVIDIA's CUDA C++ Programming guide

GPU resource management

32 threads form a warp; warps organized in blocks; blocks organized into a grid

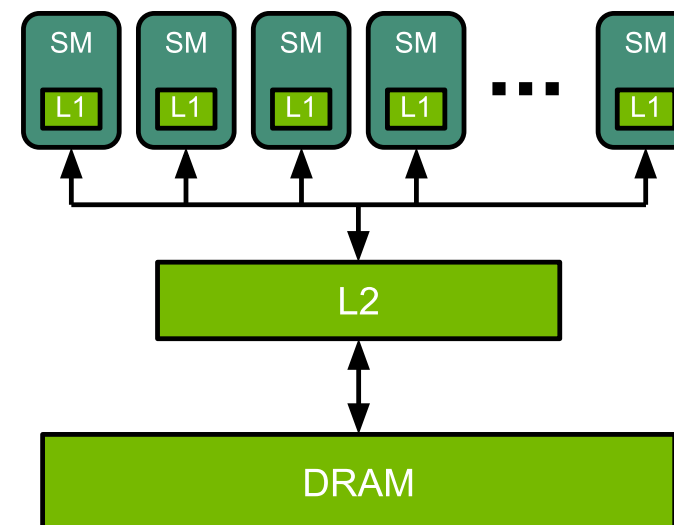
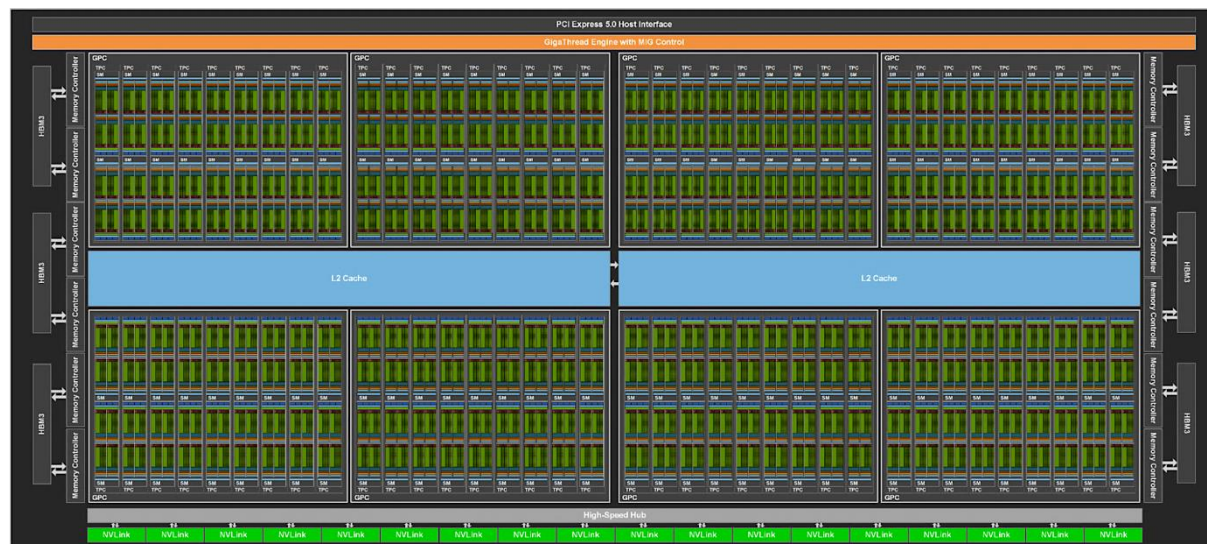
Kernel is called on a grid, resources only free up after kernel is done

Kernel launch always has overhead

Threads within a warp execute in step

No control over execution order of threads or blocks; concurrency within a warp

Multi-level of (NVIDIA) GPU organization



Physical Components

GPU card
Streaming Multiprocessors
Streaming Processors
CUDA cores / Tensor Cores

Memory

Global Memory (DRAM)
L2 Data Cache
L1 Data Caches
Registers

<https://developer.nvidia.com/blog/nvidia-hopper-architecture-in-depth/>

<https://docs.nvidia.com/deeplearning/performance/dl-performance-gpu-background/index.html#gpu-arch>

Streaming Multiprocessor (SM)

- Memory in SM:
 - L1 Data Cache / Shared Memory
- Divided into Streaming Processors (SP)
 - CUDA cores and Tensor core
 - Registers
 - Load/Store Units
 - Special Function Unit (cos(x), sin(x), e^x)



What different numbers mean

- **Compute Capability (CC)** - defines what features and instructions are available for a specific GPU architecture.
- *See CUDA Programming guide 17.2. Features and Technical Specifications (Tables 27 & 28)*
 - Higher CC means more features.
 - Tesla (CC 1.x) → Hopper (9.0) → Blackwell 12.0
 - Example: Tensor cores available from CC 7.0+, thread block clusters 9.0+*CUDA*
- CUDA Toolkit (12.9.0 as of May 2025)

Feature	A100 80GB SXM	H100 80GB SXM
GPU Architecture	Ampere	Hopper
GPU Memory	80GB HBM2e	80GB HBM3
Memory Bandwidth	2039 GB/s	3352 GB/s
L2 Cache	40MB	50MB
FP64 Performance	9.7 TFLOPS	33.5 TFLOPS
FP32 Performance	19.5 TFLOPS	66.9 TFLOPS
RT Cores	N/A	N/A
TF32 Tensor Core	312 TFLOPS	989 TFLOPS
FP16/BF16 Tensor Core	624 TFLOPS	1979 TFLOPS
FP8 Tensor Core	N/A	3958 TFLOPS
INT8 Tensor Core	1248 TOPS	3958 TOPS

<https://www.horizoniq.com/blog/h100-vs-a100-vs-l40s/>

Tensor cores

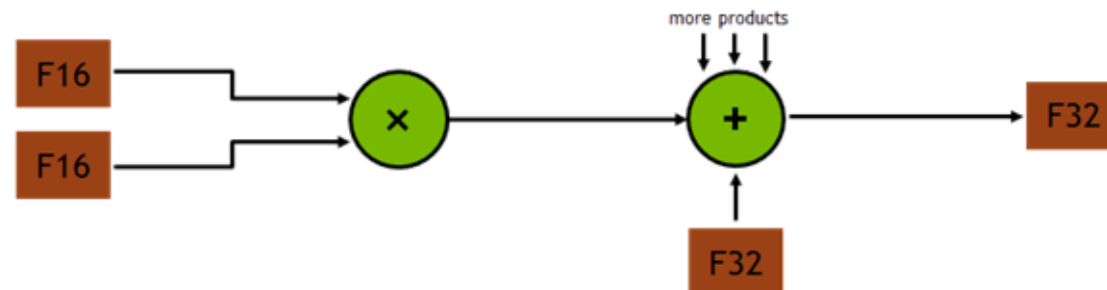
- Specialized cores operate on 4x4 matrices
- Performed matrix multiply and accumulate
- Used

$$\mathbf{D} = \begin{pmatrix} A_{0,0} & A_{0,1} & A_{0,2} & A_{0,3} \\ A_{1,0} & A_{1,1} & A_{1,2} & A_{1,3} \\ A_{2,0} & A_{2,1} & A_{2,2} & A_{2,3} \\ A_{3,0} & A_{3,1} & A_{3,2} & A_{3,3} \end{pmatrix} \begin{pmatrix} B_{0,0} & B_{0,1} & B_{0,2} & B_{0,3} \\ B_{1,0} & B_{1,1} & B_{1,2} & B_{1,3} \\ B_{2,0} & B_{2,1} & B_{2,2} & B_{2,3} \\ B_{3,0} & B_{3,1} & B_{3,2} & B_{3,3} \end{pmatrix} + \begin{pmatrix} C_{0,0} & C_{0,1} & C_{0,2} & C_{0,3} \\ C_{1,0} & C_{1,1} & C_{1,2} & C_{1,3} \\ C_{2,0} & C_{2,1} & C_{2,2} & C_{2,3} \\ C_{3,0} & C_{3,1} & C_{3,2} & C_{3,3} \end{pmatrix}$$

FP16 or FP32 FP16 FP16 FP16 or FP32

*From Nvidia Technical Blog
Programming Tensor Cores in
CUDA 9*

FP16 storage/input Full precision product Sum with FP32 accumulator Convert to FP32 result



From Nvidia Technical Blog
Programming Tensor Cores in
CUDA 9

cuBLAS Mixed-Precision GEMM
(FP16 Input, FP32 Compute)



Useful resources

- **CUDA documentation** <https://docs.nvidia.com/cuda/>
- **NVIDIA Developer forums**
<https://forums.developer.nvidia.com/c/developer-tools/106>
- **Stack Overflow** - its likely someone already had the same programming question as you and it has already been answered in the comments. (*Unlike reddit*) Stack Overflow has is mostly high quality, polite, informative answers.
- You can ask me, I will try my best to answer your question. You can also talk to your classmates, this way both of you learn.
- **Code:** https://github.com/zuzanna-m-j/IntroToCUDA_S2026.git

Login to Frontera and request a GPU node

```
ssh your_user_name@frontera.tacc.utexas.edu
```

```
idev -p rtx
```

```
module load cuda
```

```
git clone https://github.com/zuzanna-m-j/IntroToCUDA_S2026.git
```

```
cd IntroToCUDA_S2026
```

```
cd 0_Query
```

```
nvcc -o query query.cu
```

Query the device properties

- In the *0_Query* directory, execute *./query*

```
c196-082[rtx](358)$ ./query
```

```
* You have 4 CUDA-enabled devices *
```

```
=== Information for device number 0 ===
```

```
Device name: Quadro RTX 5000
```

```
Compute Capability: 7.5
```

```
Maximum threads per SM: 1024
```

```
Maximum threads per block (blockSize): 1024
```

```
Max thread block size in x:1024, y:1024, z:64
```

```
Max grid size in x:2147483647, y:65535, z:65535
```

```
Total Global Memory: 16 GB
```

```
Peak Memory Bandwidth: 448 GB/s
```

```
int numDevices;

cudaGetDeviceCount(&numDevices);

printf("\n\n * You have %d CUDA-enabled devices *\n\n", numDevices);

//enumerate though devices
for (int i = 0; i < numDevices; i++) {

    cudaDeviceProp properties;
    cudaGetDeviceProperties(&properties, i); //puts all properties in the properties variable
    printf(" === Information for device number %d ===\n", i);

    //Properties can be accessed by their name

    printf(" Device name: %s\n", properties.name);
    printf(" Compute Capability: %d.%d\n", properties.major, properties.minor);
    printf(" Maximum threads per SM: %d\n", properties.maxThreadsPerMultiProcessor);

}
```

Using nvidia-smi and property query

```
c196-082[rtx](350)$ nvidia-smi
Wed Apr 1 16:49:24 2026
```

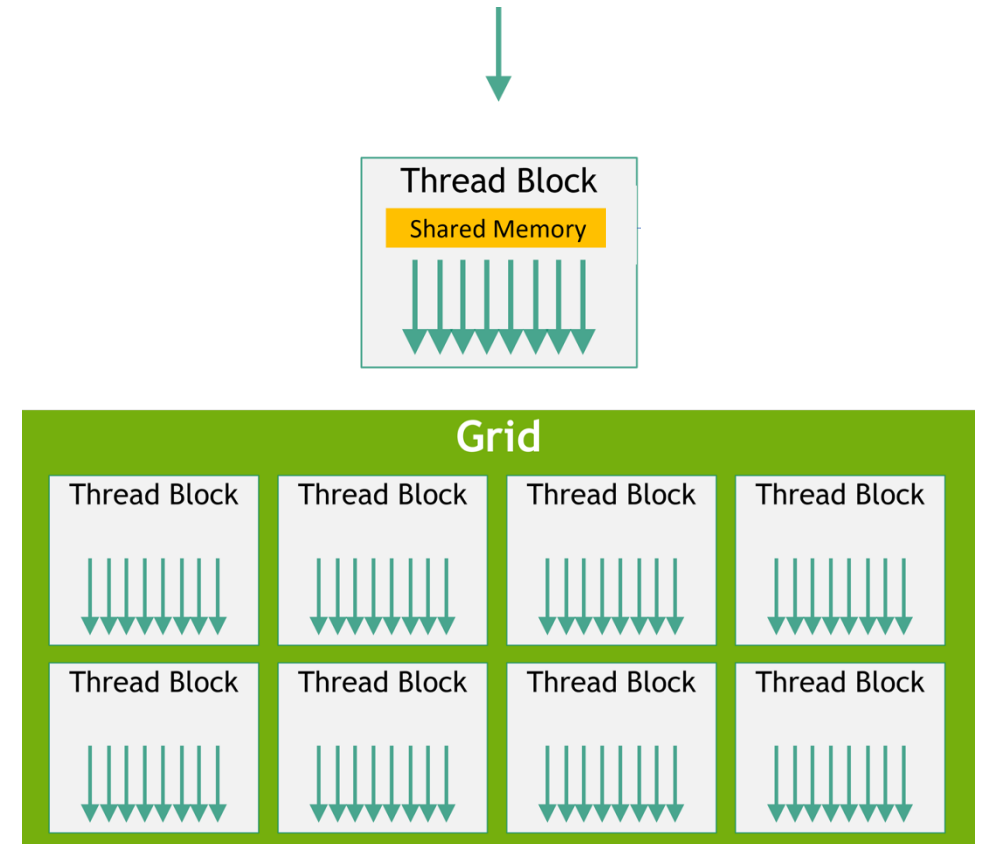
NVIDIA-SMI 535.113.01				Driver Version: 535.113.01				CUDA Version: 12.2			
GPU		Name		Persistence-M		Bus-Id		Disp.A		Volatile Uncorr. ECC	
Fan	Temp	Perf		Pwr:Usage/Cap				Memory-Usage		GPU-Util	Compute M. MIG M.
0	Quadro RTX 5000			On		00000000:02:00.0	Off				Off
0%	31C	P8		11W / 230W		0MiB / 16384MiB				0%	Default N/A
1	Quadro RTX 5000			On		00000000:03:00.0	Off				Off
0%	31C	P8		2W / 230W		0MiB / 16384MiB				0%	Default N/A
2	Quadro RTX 5000			On		00000000:82:00.0	Off				Off
0%	31C	P8		5W / 230W		0MiB / 16384MiB				0%	Default N/A
3	Quadro RTX 5000			On		00000000:83:00.0	Off				Off
0%	31C	P8		6W / 230W		0MiB / 16384MiB				0%	Default N/A
Processes:											
GPU	GI	CI	PID	Type	Process name						GPU Memory Usage
	ID	ID									
No running processes found											

Execution Model & Hardware Mapping

Threads in a warp execute in-step

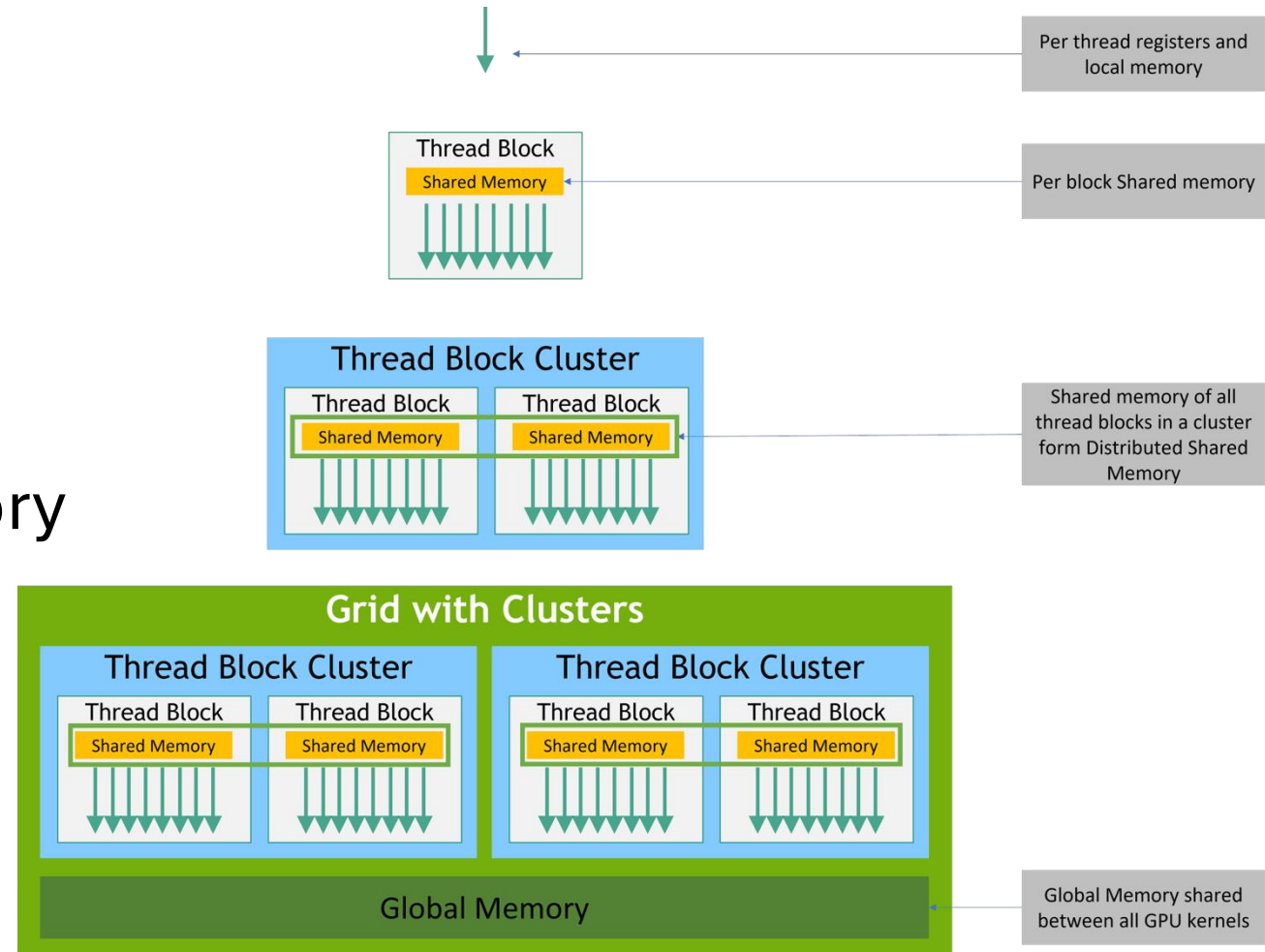
Only guarantee of warp concurrency

- Threads are mapped to CUDA cores
- Blocks mapped to SMs
- Grid mapped to the GPU



Memory Hierarchy

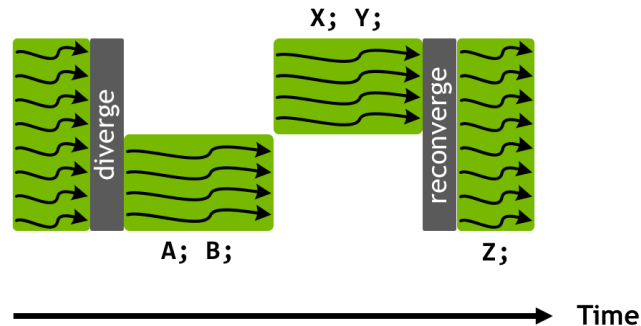
- High latency in data transfer hidden with multiple levels of memory
- Registers
- L1 local cache / Shared Memory
- L2 shared by all SMs – intermediate in Host communication
- Global Memory (DRAM)



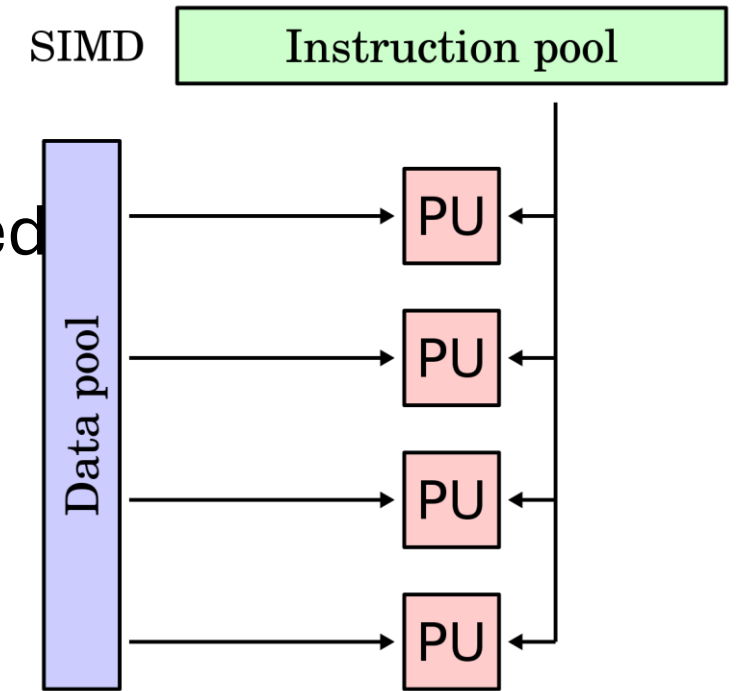
Warps and SIMT Execution Pipeline

- Threads in a warp execute in-step
- Only guarantee of warp concurrency
- Divergent branches (if; else) should be avoided

```
if (groupThreadID.x < 4){  
    A;  
    B;  
} else {  
    X;  
    Y;  
}  
Z;
```



From Nvidia Technical Blog



PU – Processing Unit, here a GPU thread

<https://en.algorithmica.org/hpc/simd/>

Migrating the data

- Data is initialized on the CPU, then copied to the GPU
- CPU $\leftrightarrow$ GPU transfers are slow, keep them to the minimum
- Ways to transfer the data:
 - Allocate space on the GPU, copy from CPU to GPU (and back)
 - Use Unified (Managed) Memory

Data allocation and movement

allocate memory on the device

```
cudaMalloc(device_pointer, size)
```

copy data between host and device

```
cudaMemcpy(dest_ptr, src_ptr, size, direction)
```

Data copy directions

0: cudaMemcpyHostToHost

1: cudaMemcpyHostToDevice

2: cudaMemcpyDeviceToHost

3: cudaMemcpyDeviceToDevice

```
int *array, *d_array;
```

```
cudaMalloc(&d_array, N * sizeof(int));
```

```
cudaMemcpy(d_array, array, N*sizeof(int), cudaMemcpyHostToDevice);
```

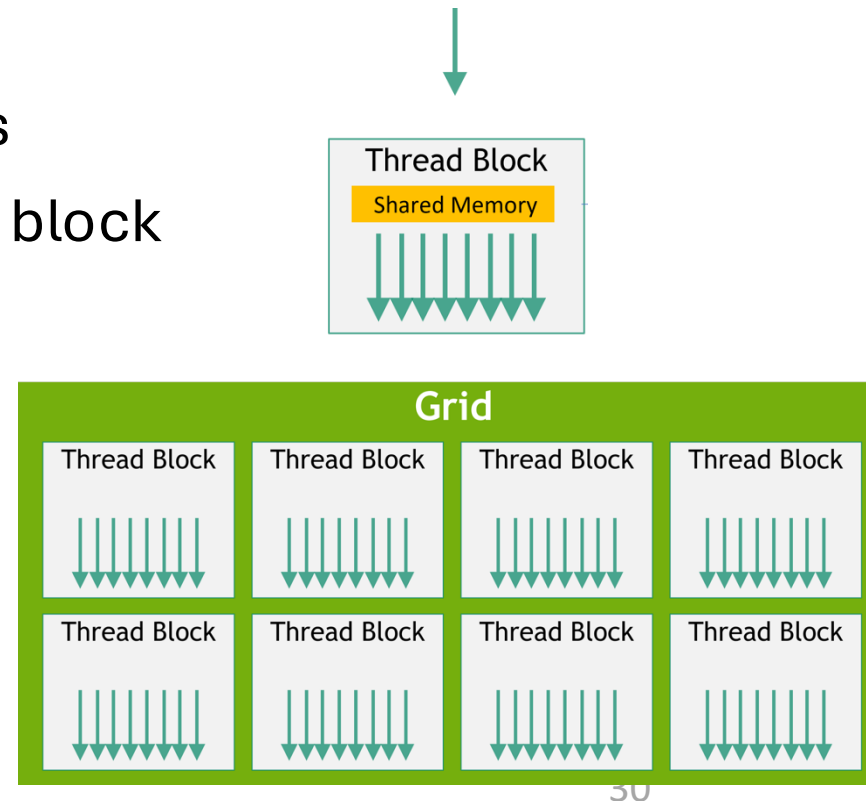
Kernel – a GPU function

- Kernels declared as normal functions with `__global__` attribute
- `__global__ myKernel( kernel parameters ){ ...kernel body...};`
- `__global__` makes the function callable from host and device
- Called with information about # of blocks and threads/block
- `myKernel<<< # Blocks , Threads per block >>>( parameters );`

Kernel configuration

- Chevron syntax <<< >>> specifies the resources provided to the kernel
- Two required positional parameters <<<**GridSize, BlockSize**>>>

- **GridSize** = Number of thread blocks
- **BlockSize** = Number of threads in a block



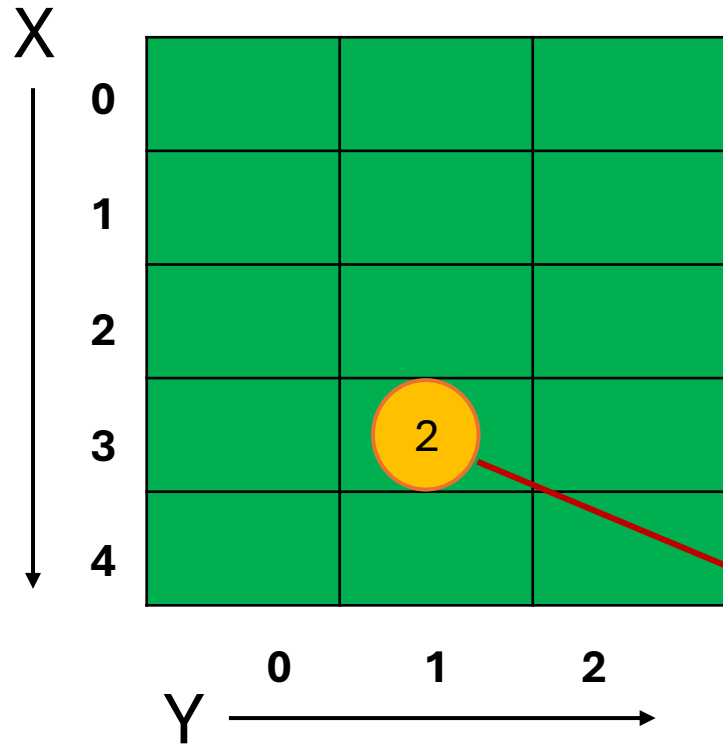
Kernel configuration

- Two-tier hierarchy:
 - **Grid** of blocks
 - **Block** of threads
- Why the division?
 - Entire block assigned to one SM – shared memory, synchronization
 - Grid specifies how many blocks we want to launch
 - Since block is mapped to SM, it will only start when resources are available

Gridsize and blocksize

- Grid size and block size are dim3 (x, y, z): can be 1D, 2D or 3D
- Block size limited by max number of threads 1024. Block size must be the multiple of 32 (warp size).
- Grid size imposes limit in each direction, no limit for total grid size (x: $2^{31}-1$, y: 65535, z: 65535)
- Multidimensional grid helps to map to multidimensional data

Indexing the threads



```
dim3 GridSize(5,3) dim3 BlockSize(512)  
someKernel<<<GridSize,BlockSize>>>();
```

Position on the grid = $\text{GridSize.x} * \text{GridIdx.y} + \text{BlockIdx.x} \rightarrow 5 * 1 + 3 = 8$

Position in the block = ThreadId.x

Global Id = $(\text{GridSize.x} * \text{GridIdx.y} + \text{BlockIdx.x}) * \text{BlockSize.x} + \text{ThreadId.x}$



Converting CPU code to GPU code

A simple serial CPU function to square an array

Loop over each elements, each iteration squares the value at array[i] and writes it back to a[i]

```
void squareArray(int* array, int N){  
  
    for (int i=0; i < N; i++){  
        array[i] = array[i] * array[i];  
    }  
}
```

Mapping threads to loop iterations

```
void squareArray(int* array, int N){  
    for (int i=0; i < N; i++){  
        array[i] = array[i] * array[i];  
    }  
}
```

```
__global__ d_squareArray(int* d_array, int N){  
    int tid = BlockSize.x * BlockIdx.x + ThreadIdx.x;  
  
    // don't go out of bounds  
    if (tid < N){  
        // each thread maps to a loop iteration  
        d_array[tid] = d_array[tid] * d_array[tid];  
    }  
}
```


Data allocation and movement

```
//host allocation and initialization
```

```
const int N = 1024;
```

```
int *array = (int *)malloc(sizeof(int) * N);
```

```
for (int i = 0; i < N; i++) {array[i] = i;}
```

```
//device allocation and copy
```

```
int *d_array;
```

```
cudaMalloc(&d_array, N * sizeof(float));
```

```
cudaMemcpy(d_array, array, N*sizeof(int), cudaMemcpyHostToDevice);
```

Running the function / kernel

```
// CPU
squareArray(array, N);

int blockSize = 512;

int gridSize = N/blockSize;

//GPU
d_squareArray <<<gridSize, blockSize>>>(d_array, N);

cudaMemcpy(array, d_array, N*sizeof(int), cudaMemcpyDeviceToHost);

//deallocate the memory
free(array);

cudaFree(d_array);
```

Kernel structure

```
__global__ void arrayAdd(float *d_a, float *d_b, float *d_out, int size)
{
    // Calculate the global thread ID

    int tid = blockIdx.x * blockDim.x + threadIdx.x;

    // Make sure we don't go out of bounds

    if (tid < size) {
        d_out[tid] = d_a[tid] + d_b[tid];
    }
}
```

Hands on exercise no. 1

Source code in [1_FirstKernel/arrayAdd.cu](#)

Simple program which:

- Initializes data on the CPU
- Copies it to GPU
- Adds arrays together
- Measures execution time of the kernel
- Copies the result back to CPU
- Measures bandwidth of the data transfers
- Shows how to perform CUDA error checking

Check for errors

```
cudaError_t err;  
  
err = cudaGetLastError();  
if (err != cudaSuccess)  
{  
    printf("CUDA Error:%s\n", cudaGetErrorString(err));  
    return -1;  
}
```

CUDA events

```
cudaEvent_t start, stop;  
cudaEventCreate(&start);  
cudaEventCreate(&stop);
```

```
cudaEventRecord(start);
```

... GPU code to time ...

```
cudaEventRecord(stop);  
cudaEventSynchronize(stop);  
cudaEventElapsedTime(&time, start, stop);
```

- More precise than CPU timers for measuring GPU execution time
- `cudaEventRecord` puts a time stamp into a given event
- `cudaEventElapsedTime` measures time between the events in ms.

Sample output

- Effective bandwidth: (total bytes read + written)/execution time
- Notice the warm-up run before the actual timing

```
./arrayAdd  
Copy H->D took 0.144992 ms  
CUDA kernel launch with 40 blocks of 256 threads  
Kernel warm up time: 36.640991 ms  
Matrix addition took 0.004992 ms  
Effective Bandwidth (GB/s): 24.038462  
Copy D->H took 0.015424 ms
```

Exercise #1: Using grid-stride loop

- The size of the input array can be larger than the number of available threads
- Threads can be reused to process multiple elements
- Grid-stride loop ensures coalesced memory

```
for (int i = blockIdx.x * blockDim.x + threadIdx.x;  
     i < size;  
     i += blockDim.x * gridDim.x)
```


Profiling tools

- For more insight we can use profiling:
- **Nsight Systems** – for profiling entire code
- **Nsight Compute** – profiling specific kernel, detailed report
- Repots can be viewed in GUI on any system
- <https://developer.nvidia.com/nsight-systems/get-started>
- <https://developer.nvidia.com/nsight-compute>

Using Nsight Systems

```
nsys profile -o arrayAddprof --stats true ./arrayAdd
```

- The command will profile the code, create arrayAddprof.nsys-rep report file and print statistics to the terminal.
- To display them again from the report, use:

```
nsys stats arrayAddprof.sqlite
```

Example output

[6/8] Executing 'cuda_gpu_kern_sum' stats report

Time (%)	Total Time (ns)	Instances	Avg (ns)	Med (ns)	Min (ns)	Max (ns)	StdDev (ns)	Name
100.0	3,360	2	1,680.0	1,680.0	1,664	1,696	22.6	arrayAdd(int *, int *, int *, int)

[7/8] Executing 'cuda_gpu_mem_time_sum' stats report

Time (%)	Total Time (ns)	Count	Avg (ns)	Med (ns)	Min (ns)	Max (ns)	StdDev (ns)	Operation
70.1	6,080	2	3,040.0	3,040.0	2,912	3,168	181.0	[CUDA memcpy Host-to-Device]
29.9	2,592	1	2,592.0	2,592.0	2,592	2,592	0.0	[CUDA memcpy Device-to-Host]

[8/8] Executing 'cuda_gpu_mem_size_sum' stats report

Total (MB)	Count	Avg (MB)	Med (MB)	Min (MB)	Max (MB)	StdDev (MB)	Operation
0.080	2	0.040	0.040	0.040	0.040	0.000	[CUDA memcpy Host-to-Device]
0.040	1	0.040	0.040	0.040	0.040	0.000	[CUDA memcpy Device-to-Host]

Using Nsight Compute

Command below tells Nsight Compute to profile `arrayAdd` kernel in the executable `arrayAdd`. Report will be saved in **`arrayAddprof.ncu-rep`**. Report name is specified with `-o` option and the `.ncu-rep` extension added automatically.

`--set` option specifies which metrics to collect.

```
ncu -o arrayAddprof --set full --kernel-name=arrayAdd ./arrayAdd
```

- Use `--import` mode to display report details in the terminal

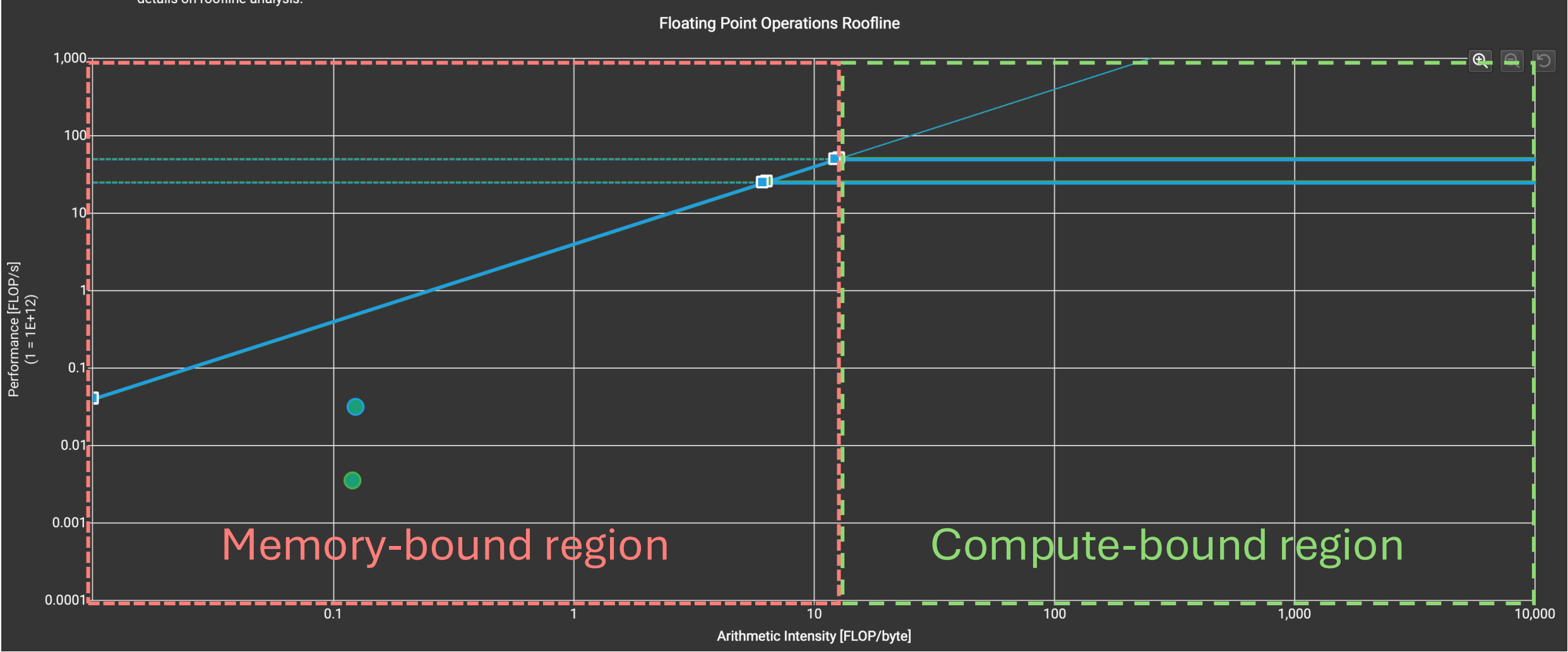
```
ncu --import arrayAddprof.ncu-rep
```

Section: Memory Workload Analysis

Metric Name	Metric Unit	Metric Value
Memory Throughput	Gbyte/s	25.17
Mem Busy	%	0.66
Max Bandwidth	%	0.63
L1/TEX Hit Rate	%	0
L2 Compression Success Rate	%	0
L2 Compression Ratio		0
L2 Hit Rate	%	36.81
Mem Pipes Busy	%	1.06

Section: GPU Speed Of Light Throughput

Metric Name	Metric Unit	Metric Value
DRAM Frequency	Ghz	2.58
SM Frequency	Ghz	1.51
Elapsed Cycles	cycle	4,298
Memory Throughput	%	0.77
DRAM Throughput	%	0.73
Duration	us	2.85
L1/TEX Cache Throughput	%	2.14
L2 Cache Throughput	%	1.50
SM Active Cycles	cycle	662.30
Compute (SM) Throughput	%	1.09



Homework #1

- You are given input array of size $2*N$ and two output arrays of size N .
- Input: *in_data*; output *out_data*, *out_tid*
- Each thread adds together two consecutive elements of *in_data* and writes the result to *out_data*. Elements added by threads do not overlap (if one thread added element 0 and 1, next thread will add elements 2 and 3). Threads are assigned to elements sequentially, starting with the thread with the lowest global ID.
- If the size of *in_data* is larger than total number of threads, use grid-size stride and start the sequence again from tid 0.
- Use the kernel configuration and N provided in `cudaHW1.cu`

Example with kernel configuration <<<1,32>>> and N = 96

```
tid = 0 : out_data[0] = in_data[0] + in_data[1]; out_tid[0] = 0
tid = 1 : out_data[1] = in_data[2] + in_data[3]; out_tid[1] = 1
tid = 2 : out_data[2] = in_data[4] + in_data[5]; out_tid[2] = 2
.
.
.
tid = 31 : out_data[31] = in_data[62] + in_data[63]; out_tid[31] = 31
tid = 0 : out_data[32] = in_data[64] + in_data[65]; out_tid[32] = 0
tid = 1 : out_data[33] = in_data[66] + in_data[67]; out_tid[33] = 1
```

Output values should be $\text{out_data}[i] == 4 * i + 1$

Index of
in_data[n]

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---

Values of
in_data[n]

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---

tid = 0

tid = 1

tid = 2

tid = 3

tid = 4

Values of
out_data[n]

1	5	9	13	17
---	---	---	----	----

Index of
out_data[n]

0	1	2	3	4
---	---	---	---	---

Values of
out_tid[n]

0	1	2	3	4
---	---	---	---	---